Banking Loan Risk Analysis Application

REST API Reference Manual & Schema Documentation

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1. Overview & Authentication

The FastAPI microservice is deployed at /api/v1. Protected endpoints require a Bearer JWT token in the Authorization header.

- **POST /api/v1/auth/login:** Authenticates user and issues access token. Form parameters: username, password.
- **GET /api/v1/auth/me:** Returns active user profile, username, and assigned role.

2. Customer Portfolio Management

- **GET /api/v1/customers:** Retrieves paginated customer records. Query params: page, page_size, risk_band, education, age_group, gender, search_id, sort_by, order.
- **GET /api/v1/customers/{id}:** Retrieves demographic details and credit risk profile.
- **GET /api/v1/customers/{id}/details:** Full customer record with 6-month historical billing statements and repayments.
- **POST /api/v1/customers:** Creates new customer profile (Administrator only).
- **PUT /api/v1/customers/{id}:** Updates customer demographic attributes.
- **DELETE /api/v1/customers/{id}:** Deletes customer and cascades to child ledgers (Administrator only).
- **GET /api/v1/customers/export/csv:** Streams customer portfolio records in CSV format.

3. Risk Analysis & Underwriting Workflow

- **POST /api/v1/risk/evaluate:** Evaluates financial parameters, returning risk score, Low/Medium/High band, and rule triggers.
- **GET /api/v1/risk/review-queue:** Fetches accounts flagged for manual review with status filters (PENDING, APPROVED, REJECTED).
- **PUT /api/v1/risk/review-queue/{customer_id}/status:** Updates underwriter decision (APPROVED / REJECTED) with audit notes.

4. Portfolio Analytics & Trends

- **GET /api/v1/analytics/kpis:** Total portfolio credit exposure, average limit, default rate, and high-risk percentage.
- **GET /api/v1/analytics/segments:** Segment KPIs grouped by Education, Age bracket, Marital status, and Gender.
- **GET /api/v1/analytics/trends:** 6-month historical billing statements, cash repayments, and delinquency rates.

5. ETL Pipeline & Batch Management

- **GET /api/v1/etl/batches:** Lists pending raw files in data/raw/ and archived files in data/processed/.
- **POST /api/v1/etl/run:** Triggers ETL on a batch from data/raw/, loads MySQL, and moves the file to data/processed/ (Administrator only).
- **GET /api/v1/etl/reconciliation:** Returns cryptographic MD5 reconciliation audit logs.
- **GET /api/v1/etl/data-quality:** Returns data quality check logs.